

UNIT-3

LEETCODEEXERCISE

1.Combine Two Tables

Problem Summary

Table: Person

Column Name	Type
PersonId	int
FirstName	varchar
LastName	varchar

PersonId is the primary key column for this table.

Table: Address

Column Name	Type
AddressId	int
personId	int
city	varchar
state	varchar

AddressId is the primary key column for this table.

Task

Write a SQL query for a report that provides the following information for each person in the Person table,

regardless if there is an address for each of those people

Solution

MySQL   Auto

```
1 # Write your MySQL query statement below
2 select FirstName, LastName, City, State
3 from Person left join Address
4 on Person.PersonId = Address.PersonId
```

Output

FirstName	LastName	City	State
Allen	Wang	null	null
Bob	Alice	New York City	New York

Expected

firstName	lastName	city	state
Allen	Wang	null	null
Bob	Alice	New York City	New York

2. Employees Earning More Than Their Managers

Problem Summary

The Employee table holds all employees including their managers.

Id	Name	Salary	ManagerId
1	Joe	70000	3
2	Henry	80000	4
3	Sam	60000	NULL
4	Max	90000	NULL

Joe is the only employee who earns more than his manager.

Employee
joe

Task

Write a SQL query that finds out employees who earn more than their managers. For the above table,

Joe is the only employee who earns more than his manager.

Solution

MySQL   Auto

```
1 # Write your MySQL query statement below
2 select a.name as employee
3 from Employee a
4 left join Employee b
5 on (a.managerid = b.id)
6 where (a.salary > b.salary)
7
```

Output

```
| employee |
|-----|
| Joe      |
```

Expected

```
| Employee |
|-----|
| Joe      |
```

3. Customer Who Visited but Did Not Make Any Transactions

Problem Summary

Table: Visits

Column Name	Type
visit_id	int
customer_id	int

visit_id is the column with unique values for this table.

This table contains information about the customers who visited the mall.

Table: Transactions

Column Name	Type
transaction_id	int
visit_id	int
amount	int

transaction_id is column with unique values for this table

Task

Write a solution to find the IDs of the users who visited without making any transactions and the number of times they made these types of visits.

Solution

MySQL   Auto

```

1 # Write your MySQL query statement below
2 SELECT
3     v.customer_id,
4     COUNT(*) AS count_no_trans
5 FROM Visits v
6 LEFT JOIN Transactions t
7     ON v.visit_id = t.visit_id
8 WHERE t.transaction_id IS NULL
9 GROUP BY v.customer_id;

```

Output

customer_id	count_no_trans
30	1
96	1
54	2

Expected

customer_id	count_no_trans
30	1
96	1
54	2

4. Rising Temperature

Problem Summary

Table: Weather

Column Name	Type
id	int
recordDate	date
temperature	int

id is the primary key for this table.

This table contains information about the temperature in a certain day.

Task

Write an SQL query to find all dates id with higher temperature compared to its previous dates (yesterday)

Solution

MySQL   Auto

```
1 # Write your MySQL query statement below
2 select w1.id
3 from Weather w1, Weather w2
4 where (w1.Temperature > w2.Temperature and
5        datediff(w1.recordDate, w2.recordDate) = 1)
```

Output

```
| id |
| -- |
| 2  |
| 4  |
```

Expected

```
| Id |
| -- |
| 2  |
| 4  |
```

5. Managers with at Least 5 Direct Reports

Problem Summary

Table: Employee

Column Name	Type
id	int
name	varchar
department	varchar
managerId	int

id is the primary key (column with unique values) for this table.

Task

Write a solution to find managers with at least five direct reports.

Solution

MySQL   Auto

```
1 # Write your MySQL query statement below
2 SELECT name
3 FROM Employee
4 WHERE id IN
5     (SELECT managerid
6      FROM Employee
7      GROUP BY managerid
8      HAVING COUNT(id) >= 5)
```

Output

```
| name |
| ---- |
| John |
```

Expected

```
| name |
| ---- |
| John |
```